

Simulating Difficult Client Behaviors in Counseling Roleplay via Activation Steering

Anonymous ACL submission

Abstract

Counselor-training systems need simulated clients that can remain case-faithful while exhibiting difficult-client behaviors such as *defensiveness*, *resistance*, *reactivity*, or *resignation*. Standard instruction-tuned LLMs tend to be overly cooperative, limiting their usefulness for such practice scenarios. We audit whether contrastive activation steering can provide runtime control over these behaviors in German counseling roleplay across four open multilingual LLMs and four bipolar persona axes. Three findings emerge. (i) Steering toward the difficult pole works: on *cooperation* and *hopefulness* it is 1–11× stronger than steering toward the easier pole, with per-model net-pairwise win rates of 25–36 percentage points. (ii) Layer selection is the single biggest practical lever: empirical downstream sweeps outperform classifier-AUC layer selection in 15 of 16 cells, since linear decodability does not imply steerability. (iii) Persona-aware vectors do not improve over a global axis vector — the standard difference-of-means construction cancels the persona-specific offset. We further package these findings as a confidence-gated evaluation protocol: where all three ensemble judges agree (72% of items), the ensemble identifies a high-confidence subset that aligns substantially better with human raters; split-judge items should be routed to human inspection.

1 Introduction

Counselor training requires exposure to client behaviors that are challenging for trainees but common in practice: defensiveness, resistance to recommendations, reactive engagement, and resignation. We refer to this jointly as *training-challenging* client behavior, and use *difficult client* as a shorthand throughout the paper for the same construct. LLM-simulated clients offer a scalable substitute for human role-players in this setting (Rudolph et al., 2024; Louie et al., 2024; Qiu

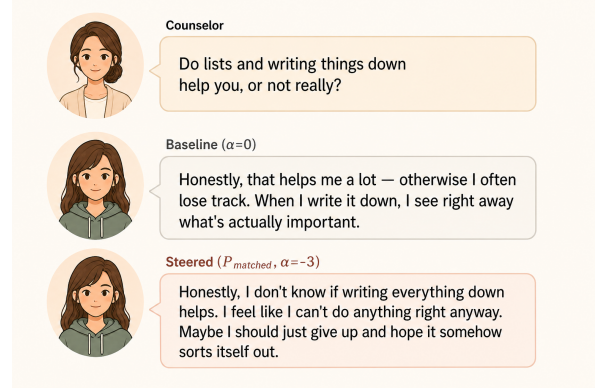


Figure 1: Activation steering toward the difficult pole. Same context, same persona (Luisa, 16, parental-conflict counseling), same model (Qwen3.5-9B *hopefulness* L15). The steering vector is added to the residual at layer 15 with $\alpha=-3$, shifting Luisa’s affective stance from constructive (baseline) to clearly *resigned* (steered) without breaking the case.

and Lan, 2024; Wang et al., 2024), but a comparative study shows they disclose more readily and resist therapeutic framing less than human role-players, leaving trainees underexposed precisely to the difficult interactions training most needs to address (Rudolph et al., 2025). The operational question for counselor-training systems is therefore not whether an LLM client can be made open, hopeful, and cooperative — it largely is by default — but whether it can be reliably steered toward the *difficult* pole of each axis without breaking the case.

We audit whether activation steering (Zou et al., 2023; Rimsky et al., 2024; Turner et al., 2025; Wu et al., 2025; Chen et al., 2025a) delivers this. Our primary evaluation steers *toward* the difficult pole at $\alpha = -3.0$ (α scales the contrastive axis-direction vector added to the residual stream at inference; negative α targets the difficult pole, positive the easier one; full method in §3): defensive openness, reactive initiative, resistant cooperation, resigned hopefulness. We additionally evaluate the

easier pole ($\alpha = +3.0$) as a symmetric control. The relevant success criterion is whether the model can be steered toward the difficult pole *without distorting the content it is supposed to be continuing* — across models, axes, and steering directions.

We make two contributions. **First**, we provide a case-grounded evaluation of activation steering for difficult-client simulation in German counseling roleplay across four open multilingual LLMs, four bipolar persona axes, and both steering directions. Steering toward the difficult pole is the practically successful direction: on *cooperation* and *hopefulness* it is 1–11 $\times$ stronger than steering toward the easier pole, with per-model net-pairwise gains of 25–36 percentage points. **Second**, we turn this evaluation into a set of deployment-relevant audits. We show that empirical downstream layer sweeps outperform classifier-AUC layer selection in 15 of 16 cells; that persona-aware contrastive vectors do not improve over a global axis vector because the standard difference-of-means construction cancels the persona-specific offset; and that reliable measurement requires judge calibration. We therefore use a confidence-gated 3-provider judge ensemble whose unanimous decisions cover 72% of items and match human raters at Cohen’s $\kappa \geq 0.43$, while split-judge items are routed to human review. Code and data are released at <https://anonymous.4open.science/r/cops-2541/>.

2 Related Work

Patient-simulation systems for counselor training span explicit cognitive models (Wang et al., 2024), LLM-to-LLM roleplay (Qiu and Lan, 2024), behavior-labelled trajectories (Qiu et al., 2026), modular standardized-patient infrastructure (Sabour et al., 2026), and persona- or principle-governed deployments (Louie et al., 2024; Steenstra et al., 2025). We are not aware of prior work in this line that addresses *runtime* persona control via steering interventions.

Persona and style conditioning is well-studied as a training-time signal (Zhang et al., 2018; Shuster et al., 2022; Xu et al., 2022; Han et al., 2023); we treat it instead as a runtime control via per-persona steering vectors.

Representation Engineering (Zou et al., 2023) and Contrastive Activation Addition (Turner et al., 2025; Rimsky et al., 2024) steer behaviour via population-level hidden-state directions without weight updates; AxBench shows simple variants

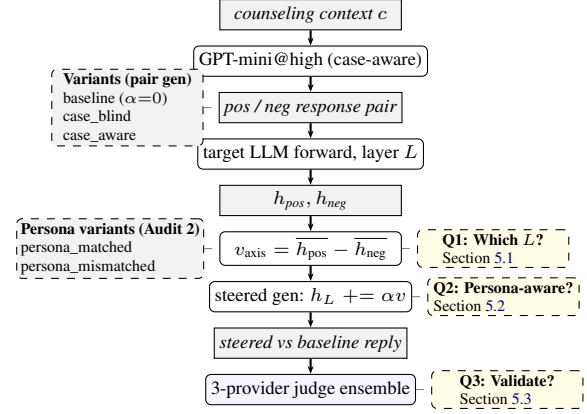


Figure 2: Pipeline (vertical) with steering-vector variants (left, dashed) and audit questions Q1–Q3 (right, dashed). Headline variants *case_blind* / *case_aware* differ in whether the pair generator sees the case context; Audit 2 contrasts per-persona vectors (*persona_matched* vs *persona_mismatched*).

remain strong against more elaborate methods (Wu et al., 2025); LoReFT learns low-rank interventions on hidden representations (Wu et al., 2024); the closest conceptual precursor is Persona Vectors (Chen et al., 2025a), which monitors and controls assistant traits via learned representation directions. LLM-as-judge reliability issues motivate our 3-provider ensemble + human validation protocol; we discuss the relevant literature in Section 3.3.

We differ from this literature in three ways. **First**, we move from generic assistant traits to role-constrained client personas in a counseling domain, on held-out conversation contexts. **Second**, we are, to our knowledge, the first to test *persona-aware* contrastive steering vectors, conditioning the steering direction not only on the stylistic axis but also on the specific client persona. **Third**, we audit the practical underpinnings of activation steering — layer choice and judge reliability — which are typically picked heuristically without empirical validation.

3 Methodology

3.1 Pipeline

Figure 2 gives an overview of the full pipeline.

For each held-in counseling context, GPT-5.4-mini@high produces two next-client utterances that differ *only* along a target axis (e.g. openness: *open* vs *defensive*). We use two generation modes that differ in whether the generator sees the case: *case_aware* includes the case description (primary concern, client profile, secondary concern) in the

generator system prompt — the default, and what Figure 2 depicts; `case_blind` omits it, so the pairs capture pure axis style without any case context. Both modes feed the same pipeline below.

For each pair, the target LLM is run forward and the residual stream at layer L (chosen per model via empirical sweep; §3.2) is averaged over the response tokens, yielding $h_{\text{pos}}, h_{\text{neg}}$.

The global axis vector is $v_{\text{axis}} = \overline{h_{\text{pos}}} - \overline{h_{\text{neg}}}$ (mean over all pairs); persona vectors $v_{\text{persona},p}$ are computed analogously over the subset of pairs for persona p .

At test time, a forward pre-hook adds $\alpha \cdot v$ to the residual stream at layer L during generation; the model receives the same chat-formatted prompt as for the unsteered baseline.

We compare five variants: a steering-free **base-line** ($\alpha = 0$); two global axis vectors using the two generation modes defined above (**case_blind** and **case_aware**); and two persona-conditioned vectors that compute $v_{\text{persona},p}$ only on pairs for persona p — **persona_matched** using the eval context’s own persona and **persona_mismatched** using a randomly drawn other persona as a control. All four steering variants apply their vector at inference time in the same way — they differ only in how the training pairs were generated.

3.2 Empirical Layer Sweep

For each (model, axis) we apply the steering vector at a single **canonical layer**, chosen empirically by sweeping candidate layers spanning the model depth and picking the one with maximum downstream net_{pw} (net-pairwise preference; defined in §3.3). The full sweep procedure and a comparison against the classifier-AUC layer-selection heuristic are reported as Audit 1 (§5.1).

3.3 Multi-Provider Judge Ensemble

Single-judge LLM-as-Judge protocols are documented to suffer from architectural self-preference (Chen et al., 2025b), position and verbosity biases (Chen et al., 2024), distributional/greedy gaps (Wang et al., 2025), and amplified non-English reliability issues (Fu and Liu, 2025). We address these with a 3-provider ensemble (different architectures), report per-stem Fleiss’s κ as a reliability flag, and validate the ensemble against human raters on a stratified sample (Section 5.3.2).

We use three judges: **Mistral-Magistral-Small-2509**, **GPT-5.4-nano@high**, and **GPT-**

OSS-120B@high. These span two architectural lineages (Mistral vs. GPT-family), limiting the risk of a single architectural self-preference dominating the winner.

Each judge receives the target axis, the polarity-aware target style label (positive or negative pole), the target instruction, the conversation context, and two randomly-ordered responses. The judge returns a 4-dimension rubric (`axis_alignment`, `case_fidelity`, `client_role_fidelity`, `training_utility`, each 1–5) and a pairwise winner $\in \{A, B, \text{tie}\}$. Full prompt text is in Appendix B.

The ensemble winner is the majority vote (defaulting to *tie* when no majority exists, i.e. all three judges disagree). Per-stem reliability is Fleiss’s κ over the three provider winner sequences. We aggregate ensemble winners per (variant, α) cell as the **net-pairwise preference**

$$net_{\text{pw}} = \frac{\# \text{steered wins} - \# \text{base wins}}{N},$$

where N is the number of paired contexts; net_{pw} ranges from -1 (base always wins) to $+1$ (steered always wins), with 0 indicating parity (ties contribute neither numerator term). We additionally report bootstrap 95% CIs on net_{pw} and paired bootstrap tests for variant comparisons (Section E).

For independent content validation we additionally classify every generated response with the German-language OnCoCo XLM-RoBERTa-large sentence classifier of Albrecht et al. (2026), fine-tuned on labeled real (non-LLM) German online counseling chat with a 28-class client utterance taxonomy. Pipeline details and results are in Section 5.3.3.

3.4 Steering Direction

The operationally relevant direction for counselor training is the *difficult* pole: the negative pole of each axis, namely defensive (openness), reactive (initiative), resistant (cooperation), and resigned (hopefulness). We make this the primary steering direction with $\alpha \in \{-1.5, -3.0\}$. We additionally evaluate the potentially easier pole at $\alpha \in \{+1.5, +3.0\}$ as a symmetric control. The target style fed to the judge is polarity-aware: positive pole label for $\alpha > 0$, negative pole label for $\alpha < 0$.

3.5 Data and Models

76 German counseling roleplay conversations collected in an elective course on online counseling

within a social work degree program, recorded between student pairs (one playing counselor, one playing client), rolling-window segmented into ≈ 5400 contexts. Each of the 10 personas has one primary concern and contributes 20–200 contrast pairs (persona-imbalanced). Held-in indices 0–199 seed pair generation; indices 400–449 form a disjoint dev split used for empirical layer selection (Audit 1, §5.1); held-out indices 200–399 (disjoint from both) drive the main evaluation. Consent, ethics review, and data-release scope are described in the [Ethics Statement](#).

Four open multilingual instruction-tuned models: Qwen3-4B (Qwen-4B), Gemma-4-E4B-it (Gemma), Qwen3.5-9B (Qwen-9B), and Mistral-7B-Instruct-v0.3 (Mistral). This spans three families (Qwen, Gemma, Mistral) and two size classes (4B vs 7–9B).

4 Headline Results: Full-Matrix Evaluation

We define a *stem* as a (model, axis, canonical-layer) configuration. With the 4 models (Qwen-4B, Gemma, Qwen-9B, Mistral; §3.5) and the 4 bipolar persona axes (openness, initiative, cooperation, hopefulness), this gives $4 \times 4 = 16$ stems. For each stem we generate the baseline ($\alpha=0$) plus 4 steering variants (case_blind, case_aware, persona_matched, persona_mismatched) $\times 2$ magnitudes in each direction ($\alpha \in \{\pm 1.5, \pm 3.0\}$) on $N=200$ held-out contexts at the canonical layer (§5.1).

4.1 Steering Toward the Difficult Pole

To avoid inflated results from per-cell variant selection, we report the prespecified net_{pw} at $\alpha=-3$ for the two case-agnostic variants (CB = case_blind; CA = case_aware) in Table 1. Both are fixed methodological choices, committed to before evaluation; a per-cell best-of-four oracle (Appendix Table 12) adds only +0.048 on the mean.

Primary finding. Steering toward the difficult pole is the strongly successful direction, and the asymmetry is consistent across all four models. The effect is clearest on *cooperation* and *hopefulness* — the two axes most relevant for counselor training: there, steering produces resistant or resigned client behavior on demand and is 1–11 $\times$ stronger across cells than the corresponding positive-pole steering on the same stems (Table 1; positive-pole comparison in Appendix Table 15;

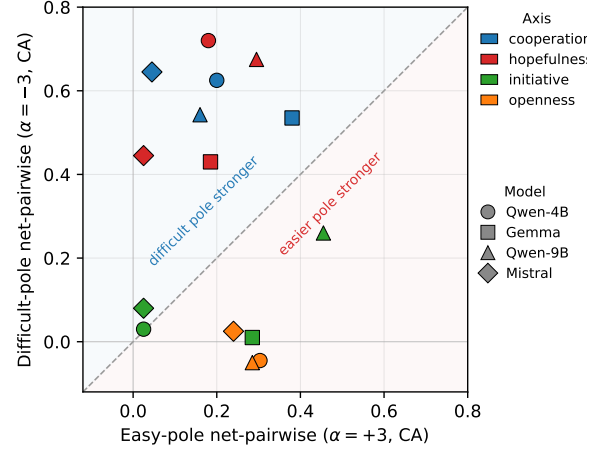


Figure 3: Asymmetric steering across the 16 paired (model, axis) cells at $|\alpha|=3$, case_aware variant. Each point plots difficult-pole net_{pw} ($\alpha=-3$) against easy-pole net_{pw} ($\alpha=+3$); points above the $y=x$ diagonal steer more strongly toward the difficult pole.

per-cell CA net_{pw} reaches +0.45–+0.72, all-cells mean +0.309, +0.234 for CB). The two remaining axes are weaker on both sides as measured by the judge ensemble: *openness/defensive* stays near zero across models, and *initiative/reactive* mirrors its weakness at the positive pole. For *defensive*, independent lexical evidence and a rubric-sensitivity ablation suggest this reflects a judge-sensitivity gap rather than a steering failure (§5.3.3, §6). No consistent size-scaling effect appears across these four models.

Instruction-tuned and RLHF-trained models are base-biased toward agreeable, cooperative, and hopeful behavior, leaving more steering headroom toward the difficult pole, consistent with prior asymmetric-steering reports: suppressing sycophancy works better than amplifying it (Rimsky et al., 2024), and ablating the refusal direction preferentially removes safety alignment rather than strengthens it (Arditi et al., 2024).

Complementarity with prompting. A 2×2 factorial ablation (persona prompt: on/off $\times$ activation steering: on/off) across the full 4×4 (model, axis) headline matrix shows a clean strong/weak axis split. On all 8 strong-axis stems (*cooperation*, *hopefulness* $\times$ 4 models) combining a negative-pole persona prompt with activation steering uniformly beats both alternatives: combining $>$ steering on 8/8 (mean lift +0.39 axis_alignment points, 1–5 scale), combining $>$ prompting on 8/8 (mean +0.84). On the 8 weak-axis stems (*defensive*, *re-*

Model	Pole	L	CB	CA	κ
Qwen-4B	defensive	15	-0.080	-0.045	0.53
Qwen-4B	reactive	9	-0.095	+0.030	0.57
Qwen-4B	resistant	20	+0.635*	+0.625*	0.48
Qwen-4B	resigned	18	+0.610*	+0.720*	0.53
Gemma	defensive	3	+0.120	+0.020	0.62
Gemma	reactive	13	-0.025	+0.010	0.52
Gemma	resistant	23	+0.285*	+0.535*	0.49
Gemma	resigned	25	+0.495*	+0.430*	0.61
Qwen-9B	defensive	20	-0.030	-0.050	0.36
Qwen-9B	reactive	20	-0.045	+0.260*	0.32
Qwen-9B	resistant	15	+0.485*	+0.543*	0.39
Qwen-9B	resigned	15	+0.510*	+0.675*	0.45
Mistral	defensive	15	-0.070	+0.025	0.37
Mistral	reactive	5	-0.020	+0.080	0.51
Mistral	resistant	15	+0.620*	+0.645*	0.39
Mistral	resigned	20	+0.352*	+0.445*	0.43
Mean			+0.234	+0.309	

Table 1: net_{pw} at $\alpha=-3$ for **CB**(case_blind) and **CA**(case_aware), no per-cell variant selection. * = 95% persona-clustered bootstrap CI excludes 0 (Table 13, App. E). All *resistant/resigned* cells significant; no *defensive/reactive* cells (except Qwen-9B *reactive* CA).

active), steering does not exceed prompting on the judge metric — consistent with the near-zero judge-measured net_{pw} and 0-overlapping CIs reported above (for *defensive*, convergent evidence in §5.3.3 suggests this understates the true effect) — yet combining still adds value over steering in 7/8 cells. Full 2×2 setup and per-cell numbers in Appendix F.

4.2 Symmetric Control: Steering Toward the Potentially Easier Pole

For symmetry we evaluate the potentially easier pole at $\alpha \in \{1.5, 3.0\}$. Averaged across the 16 (model, axis) cells, prespecified CB yields $net_{pw} = +0.195$ and CA yields $+0.206$ (Appendix J, Table 14). Twelve cells yield significant positive-pole effects (bootstrap 95% CI excludes 0) but at substantially smaller magnitudes than the difficult pole (exceptions: Qwen3-4B and Mistral-7B *initiative*, consistent with the weak-axis pattern of §4.1). Even choosing the best variant for each (model, axis) cell in hindsight (Table 15) adds only $+0.070$ over prespecified CA.

4.3 Case and Role Fidelity Under Steering

The net_{pw} metric scores overall preference but does not show *what* is traded off. Each ensemble judge also scores three non-axis 1–5 Likert dimensions (case_fidelity, client_role_fidelity, training_utility; §3.3). Aggregated across all variants and α -signs, mean $\Delta case$ and $\Delta role$

stay within $[-0.052, +0.094]$ — roughly an order of magnitude smaller than the $\Delta axis$ effect, and remaining within these bounds as $|\alpha|$ doubles from 1.5 to 3.0 (Appendix Table 19). Per-cell outliers (Appendix Table 20) concentrate on Qwen3.5-9B *hopefulness/cooperation* at $\alpha = -3.0$ under case_blind ($\Delta case -0.29, -0.27$), while case_aware and the persona-conditioned variants on the same stems hold $\Delta case$ within ± 0.05 and score higher net_{pw} . For deployment we recommend reporting net_{pw} together with $\Delta case$ and $\Delta role$ per cell — same judge call, same generations.

4.4 Single-Provider Sensitivity

Robustness check. Single-provider net_{pw} varies systematically: Mistral-Magistral runs $\sim 30\%$ lower than the GPT-family (Nano and GPT-OSS-120B agree closely; Appendix J, Table 18). Direction is preserved across providers on strong stems but can flip sign on weak ones; the bias mechanism is audited as part of Audit 3 (Section 5.3). This justifies the multi-provider ensemble.

5 Methodological Audits

Three practical choices shape the reliability of any activation-steering audit: where to inject the vector (layer selection), what to condition the vector on (axis vs. persona), and who decides whether the resulting output matches the target (judging and content validation). We report each as a self-contained audit with concrete recommendations for deployment.

5.1 Audit 1: Layer Selection — Empirical Sweep vs. Classifier AUC

For each (model, axis) we contrast two layer-selection strategies. The **AUC-best layer** is the one at which a logistic classifier separates positive-from negative-pole hidden states with highest test AUC (Zou et al., 2023; Ju et al., 2025); this heuristic has been shown to be a poor predictor of actual steering success (Billa, 2026), motivating the empirical sweep we propose here. The **sweep-best layer** is the one with maximum downstream net_{pw} (§3.3): we generate at five candidate layers spanning the model depth, with $\alpha = 1.5$ on $N = 50$ dev-split contexts (indices 400–449, disjoint from the main-eval split 200–399) under baseline+case_blind; each generation is judged by the 3-provider ensemble, and the canonical layer is the one that maximises net_{pw} .

Model	Axis	AUC L	AUC net_{pw}	Sweep L	Sweep net_{pw}
Gemma	cooperation	23	+0.12	23	+0.12
Gemma	hopefulness	13	+0.12	25	+0.32
Gemma	initiative	8	-0.06	13	+0.06
Gemma	openness	3	+0.20	3	+0.20
Mistral	cooperation	5	+0.02	15	+0.10
Mistral	hopefulness	5	+0.12	20	+0.38
Mistral	initiative	5	+0.28	5	+0.28
Mistral	openness	5	-0.08	15	+0.24
Qwen-4B	cooperation	9	+0.06	20	+0.22
Qwen-4B	hopefulness	7	-0.04	18	+0.06
Qwen-4B	initiative	9	+0.00	9	+0.00
Qwen-4B	openness	1	+0.02	15	+0.32
Qwen-9B	cooperation	5	+0.08	15	+0.20
Qwen-9B	hopefulness	5	n/a	15	+0.42
Qwen-9B	initiative	5	+0.10	20	+0.18
Qwen-9B	openness	5	+0.18	20	+0.50

Table 2: Layer selection: AUC-best vs empirical sweep-best layer. net_{pw} at $\alpha=1.5$, case_blind, $N=50$ dev-split contexts (indices 400–449, disjoint from the $N=200$ main-eval split). AUC matches sweep in only 1/16 cells.

Classifier-AUC layer selection systematically misses where steering actually works. In 15 of 16 cells the empirical sweep locates a deeper layer with substantially stronger downstream effect (Table 2). The Qwen3-4B *openness* stem illustrates the typical gap: AUC picks the first layer (classifier AUC ≈ 0.99) but yields essentially no behavior change ($net_{pw} = +0.02$), while the sweep-best layer 15 yields $net_{pw} = +0.32$ — a $16\times$ improvement.

AUC measures only whether the poles are linearly *decodable*; downstream steerability additionally requires the residual move into a region the upstream layers translate into stylistic output, and the net_{pw} gain (up to $16\times$ across cells) at modest cost (~ 40 small generation jobs at $N=50$ plus three judge passes) makes the downstream sweep the more reliable selection criterion.

5.2 Audit 2: Persona-Aware Conditioning Does Not Help

Recall that the global vector $v_{axis} = \overline{h_{pos}} - \overline{h_{neg}}$ averages over *all* contrast pairs, whereas the persona-conditioned vector $v_{persona}$ averages only over the subset of pairs belonging to persona p (§3). If persona conditioning captures meaningful persona-specific signal, persona_matched (steering with the eval context’s own persona vector) should outperform persona_mismatched (steering with a randomly drawn *other* persona’s vector) more often than chance.

Persona-conditioned vectors are statistically indistinguishable from random. We test

whether persona_matched yields higher net_{pw} than persona_mismatched in each of the 14 (model, axis) stems that have enough persona pairs. The difference is significant in only 1 of 14 stems under bootstrap 95% CI (2/14 under Wilcoxon signed-rank at $\alpha_{test} = 0.05$) — no better than the expected false-positive rate.

A second view confirms this: across the 7 Gemma/4B stems and 10 personas, each of the four steering variants (case_blind, case_aware, persona_matched, persona_mismatched) wins the best- net_{pw} comparison in $\approx 25\%$ of (stem, persona) combinations (Appendix Table 16), exactly what a uniform random assignment would produce. The ranking is also unstable across steering magnitude — in 4 of 7 stems the best-performing variant changes when α moves from 1.5 to 3.0.

Why persona conditioning might collapse to the global vector. Let $\bar{h}_{pos,p}$ and $\bar{h}_{neg,p}$ denote the mean hidden states for the positive- and negative-pole pairs of persona p , and let δ_p be the persona-specific offset — the component of the mean activation that depends on persona identity but not on axis polarity. Because δ_p is shared across both poles, it cancels in the subtraction:

$$\begin{aligned}
 v_{persona} &= \bar{h}_{pos,p} - \bar{h}_{neg,p} \\
 &= (\bar{h}_{pos} + \delta_p + \epsilon_1) - (\bar{h}_{neg} + \delta_p + \epsilon_2) \\
 &= v_{axis} + (\epsilon_1 - \epsilon_2),
 \end{aligned}$$

where ϵ_1, ϵ_2 are finite-sample noise from averaging over only persona p ’s pairs (20–200 per persona) instead of all pairs. Empirically, $\cos(v_{persona}, v_{axis}) = 0.78\text{--}0.997$ across all stems, confirming that persona vectors are near-parallel to the global vector.

We test three alternative formulations designed to isolate persona-specific signal — an orthogonal residual, a cross-persona contrast, and a direct interaction residual (definitions in Appendix J.0.1) — but none produces meaningful steering on the four 4B/Gemma stems (Appendix Table 17). At our data scale the standard persona-vector construction confounds persona-axis interaction with the bulk of v_{axis} ; larger per-persona datasets or non-linear interventions may still recover persona-specific effects.

5.3 Audit 3: Judge Reliability and Content Validation

Activation-steering effects measured by an LLM-judge ensemble are only as trustworthy as the

Rater / aggregate	n	κ vs. Ens.	Pearson r vs. Δ_{aa}
Rater 1	124	0.47	0.55
Rater 2	111	0.40	0.59
Rater 3	123	0.29	0.49
Rater 4	124	0.32	0.50
Human-maj. (joint)	110	0.36	—
↔ easy	53	0.78	—
↔ stress	28	0.17	—
↔ near tie	29	-0.08	—
Fleiss κ humans (n=110)		0.38	—
Fleiss κ judges (n=110)		0.63	—

Table 3: Inter-rater reliability headline. Cohen’s κ on 3-category (base/steered/tie) winner labels against the 3-provider ensemble winner. Human-majority breakdown by ensemble-sample stratum (easy: clear ensemble winner; near_tie: ensemble uncertainty by design). Pearson r is computed on the continuous ensemble $\Delta_{axis\text{-alignment}}$ vs. human winner coded as $\pm 1/0$.

judges agreeing on them. We audit ensemble agreement and calibration, positional bias, and human alignment, and then validate the resulting content shifts with an external sentence classifier. Full audit details (provider calibration, four-test positional-bias analysis, per-stratum, per-axis, and n_{agree} -partitioned breakdowns) are in Appendices H–I.

5.3.1 Ensemble Agreement and Calibration

The three-provider ensemble reaches substantial agreement on most stems but diverges on weak axes, where sign disagreement can mask real effects. Mistral-Magistral runs systematically lower than the GPT-family; on strong stems rankings are preserved, but on weak stems the providers can vote opposite directions (e.g. Qwen3-4B *initiative* L9). We flag low- κ stems in the full-matrix tables. A four-test positional-bias audit identifies GPT-OSS-120B as the position-debiased anchor (Appendix H). Per-stem Fleiss’s κ ranges 0.17–0.55 (median 0.45; 11/15 stems ≥ 0.4 ; Appendix Tables 7, 18).

5.3.2 Human Validation

Four human raters annotated a 124-item stratified sample (4 axes $\times$ 2 poles $\times$ 4 models $\times$ 4 variants; protocol in Appendix I). The human majority prefers the steered response on 47% of items vs. 32% for the base (21% tie), rising to 58%/34% on the *easy* stratum — corroborating the automated direction. Inter-human Fleiss’s $\kappa = 0.38$ vs. inter-judge $\kappa = 0.63$ on the same items (Table 3); the gap is consistent with shared biases among LLM judges (Gu et al., 2024; Chen et al., 2024).

The stratum-resolved view validates the ensemble where it matters. On *easy* items (clear ensemble

agreement, $|\Delta_{aa}| \geq 2$), human–ensemble Cohen’s $\kappa = 0.78$ ($n=53$); on *near_tie* items it collapses to -0.08 — ambiguous items stay ambiguous under human inspection. Partitioning by n_{agree} (judges voting the ensemble winner) shows this is mediated by judge unanimity: on $n_{agree}=3$ items (72% of the sample) every rater reaches Cohen’s $\kappa \geq 0.40$; on $n_{agree}=2$ items every rater drops to $\kappa \leq 0.19$. This justifies $n_{agree}=3$ as a high-confidence filter for the headline matrix (§4); split-judge items are routed to human review.

Qualitative inspection of the openness/defensive disagreements suggests the ensemble undercounts steering on this axis: judges weight the rubric’s self-protective/justificatory wording over its withholding/avoidant wording (§6, Appendix I).

5.3.3 Convergent Validation via OnCoCo Sentence Classifier

The judge ensemble scores axis alignment directly — leaving open whether the effect is genuine content shift or judge-perceived stylistic noise. For independent evidence we classify every generated sentence with the OnCoCo XLM-RoBERTa-large classifier (Albrecht et al., 2026), fine-tuned on (non-LLM) German counseling chat over a 28-class client-utterance taxonomy (86,716 sentences total; pipeline in Appendix L). A 28×2 χ^2 contingency test per (axis, pole) cell with cluster-robust permutation correction (Appendix Table 21, procedure in Appendix M) replicates the pattern of §4: *cooperation* and *hopefulness* yield highly significant predicted-label shifts in both pole directions ($p_{cluster} = 0.005$ on all four cells); *openness* and *initiative* do not ($p \geq 0.10$). A continuous-logit reanalysis using Jensen-Shannon divergence (JSD) on the full 28-class softmax distributions (rather than argmax labels) recovers the *openness* effect: $p_{JSD} < 0.001$ on the positive pole ($200\times$ more sensitive than argmax χ^2) and $p_{JSD} = 0.036$ on the negative pole, while *initiative* remains non-significant ($p_{JSD} \geq 0.45$). Per-label two-proportion z -tests (Appendix N) show shifts in the clinically expected direction: under *resigned*, more emotional self-disclosure and setbacks, fewer progress reports or agreements; under *resistant*, more negative feedback on recommendations, fewer agreements or formal closures. The *initiative* OnCoCo null co-occurs with a Monroe lexical signature in the expected direction (Appendix O; e.g. *defensive*: -33% emotion-word frequency on Qwen3-4B). Both nulls are consistent with steering affecting af-

factive texture — withholding, emotional flattening — rather than the content categories the OnCoCo taxonomy discriminates. For *openness*, the JSD recovery above and the Monroe emotion-word reduction together indicate that the judge-measured null reflects a measurement gap, not a steering failure (qualitative evidence in §6, Appendix I). The convergence across two independent measurement chains (judge ensemble + OnCoCo) on *cooperation* and *hopefulness* validates that the detected shifts there are genuine construct movement rather than rubric artifacts. A rubric-sensitivity ablation ($N=200$, CB, $\alpha=-3.0$, same 3-provider ensemble) that replaces *defensive* with *closed-off/withholding* raises net_{pw} from $+0.04$ to $+0.14$ on three of four models (31 base→steered vs. 21 steered→base flips), confirming rubric sensitivity as a substantial contributor to the observed null.

6 Discussion

Three operational recommendations emerge from our audits. Empirical layer sweeps yield up to $16\times$ larger net_{pw} than classifier-AUC selection across the 16 (model, axis) cells we test: linear decodability does not imply steerability. At our data scale, three alternative persona-vector constructions (orthogonal residual, cross-persona contrast, interaction term) all collapse to the global axis vector v_{axis} — the standard $v_{persona}$ construction cancels the persona offset by subtraction. And single-judge results differ from the 3-provider ensemble by $\sim 30\%$ (Mistral conservative, GPT-family liberal); reporting Fleiss’s κ per stem at $3\times$ the API cost converts this hidden noise into a reportable reliability flag.

For the operationally relevant counselor-training scenario — defensive, resistant, or resigned client on demand — $\alpha = -3.0$ steering yields effects $1\text{--}11\times$ larger than the positive direction on *cooperation* and *hopefulness*, peaking at $net_{pw} = +0.75$ on Qwen3.5-9B *resigned* (oracle; Appendix J). The asymmetry is consistent with prior reports that RLHF-tuned models are base-biased toward agreeable behavior, leaving more headroom away from this default: suppressing sycophancy works better than amplifying it (Rimsky et al., 2024), and ablating the refusal direction is straightforward (Arditi et al., 2024). For practice this is good news: the difficult pole is the direction steering delivers most reliably.

Appendix F reports the prompt-vs-steering ablation across the full 16-stem matrix: combining

beats steering on all 8 strong-axis stems and on 7/8 weak-axis stems; combining beats prompting on all 8 strong-axis stems. On the weak axes steering itself fails to exceed prompting (0/8), in line with the axis-level net_{pw} asymmetry. This is consistent with prior findings of direction-dependent strengths (Rimsky et al., 2024; Wu et al., 2025) — few-shot prompting outperforms CAA at amplifying agreeable behavior, while CAA outperforms prompting at suppressing it, precisely the direction we use. For counselor-training deployments where the target is the difficult pole on *cooperation* or *hopefulness*, combine the two rather than choose between them.

A fourth observation concerns the mechanism behind the *openness/defensive* judge null. The rubric names both withholding/avoidant (*reserved*, *evasive*) and self-protective aspects (*self-protective*, *justificatory*); in practice judges weight the latter, scoring hedge-rich baselines as more defensive than steered outputs that present a flat positive façade with little affective content (Items 89, 96, 98; Appendix I). The rubric-sensitivity ablation in §5.3.3 confirms this quantitatively: replacing *defensive* with *closed-off/withholding* recovers a positive net_{pw} on three of four models.

7 Conclusion

We audited activation steering for difficult-client simulation in counseling roleplay across four open LLMs and four bipolar persona axes. Three findings emerge. (1) Steering toward the difficult pole is the strongly successful direction: $net_{pw} = +0.75$ on Qwen3.5-9B *hopefulness* (resigned), $1\text{--}11\times$ stronger than the easy-pole direction on *cooperation* and *hopefulness*. (2) Layer selection is the single biggest practical lever: empirical downstream sweeps yield up to $16\times$ larger net_{pw} than classifier-AUC selection across the 16 (model, axis) cells we test — linear decodability does not imply steerability. (3) Persona-aware variants do not significantly outperform a global axis vector — the standard $v_{persona}$ construction structurally cancels the persona offset, and matched-vs-mismatched persona vectors differ significantly in only 1 of 14 cells. We release the pipeline for activation-steering audits on case-grounded roleplay in counseling and related simulation tasks.

Ethical Considerations

The counselling dialogue data used in this study were collected from structured role-play sessions conducted by social-work students in an elective online-counselling course of a university degree programme. Participants acted in predefined client and counsellor roles; no real clients, patients, or personal therapeutic information were involved. All participants provided informed consent for research use of the data. Participation was a component of the course practicum, but the recordings were not used to grade students; to mitigate the residual compulsion of the course-component requirement, students chose their own client persona, allowing them to avoid topics that could conflict with their own circumstances.

Anonymisation. The ten client personas (“Frau Schuster”, “Luisa”, etc.) and their case narratives are fictional and were self-designed by the student authors. In contrast, the student playing the counsellor role often introduced themselves in the roleplay with what may have been their own first name. We therefore ran an additional PII detection pass over every unique conversation turn (5 235 turns) using openai/privacy-filter (1.5 B-parameter token classifier, q4f16-ONNX, pinned revision 7ffa9a04. . . , executed locally), then manually curated the flagged surfaces. Counsellor first names, full names, and explicit surnames were replaced by [BERATER:IN]; the home city name by [STADT]; named counselling-centre identifiers by [BERATUNGSSTELLE]; URLs and emails by [URL] and [EMAIL] respectively. The audit log records 581 substitutions across the four data files. Persona names, family-member references, and DPO-training personas are kept verbatim because the steering audit depends on them.

Intended use and deployment risks. Recent conceptual work distinguishes autonomous counsellor bots, AI training simulators, and counsellor-facing augmentation tools as ethically distinct implementation approaches (Steigerwald et al., 2026a). Our work is positioned in the training-simulator setting and not as an autonomous counselling agent. The intended application — controllable difficult-client simulation for counsellor training — serves an educational purpose that may improve the quality of counselling services. Steering models toward defensiveness, resistance, or resignation can be useful pedagogically but also

risks normalising adversarial or clinically inappropriate interactions if used without instructor oversight. The system should therefore be limited to training contexts, clearly labelled as simulation, and evaluated by domain experts for case fidelity (Section 4.3), escalation behaviour, and avoidance of harmful clinical advice.

AI tool use. Generative AI tools (ChatGPT and Claude) were used throughout the article for coding assistance, literature discovery, LaTeX figure and table preparation, and clarity improvements. They were not used to generate scientific claims or experimental results. All content was reviewed, verified, and finalized by the authors.

Limitations

All empirical evaluations are conducted on German counseling roleplay dialogues; other case-grounded roleplay domains remain untested. The persona set covers $N = 10$ clients with up to 10:1 imbalance across personas, limiting per-persona statistical power. Model coverage is restricted to the 4B–9B parameter range across three families (Qwen, Gemma, Mistral), and whether findings hold for larger models is unknown. The prompt-conditioning comparison (Appendix F) is single-provider (GPT-OSS-120B only, the position-debiased anchor from §5.3) rather than the 3-provider ensemble used for the headline matrix; this is a trade-off chosen to suppress the small-effect noise that Mistral-Magistral’s position bias introduces on the weak axes. No comparison against learned-intervention methods such as LoReFT (Wu et al., 2024) is included. Human evaluation confirms the steering effect on a 110-item stratified sample (§5.3.2) but does not assess response quality beyond persona-match preference; whether steered responses remain clinically appropriate is not rated.

Each of our 10 personas has exactly one case, so the contrastive pairs entangle persona and case; the v_{persona} collapse reported in Audit 2 (Section 5.2) may not hold in domains with multiple cases per persona, where the pair construction decouples the two more cleanly.

We steer one axis at a time. Whether two steering vectors $\alpha_1 v_1 + \alpha_2 v_2$ deliver compositional personas linearly, with interference, or with non-trivial geometric interactions is an open question that our pipeline supports but does not validate.

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	A Persona Axis Definitions	938															
	<table><tr><th>Axis</th><th>Positive pole</th><th>Negative pole</th></tr><tr><td>openness</td><td>open</td><td>defensive</td></tr><tr><td>initiative</td><td>explorative</td><td>reactive</td></tr><tr><td>cooperation</td><td>cooperative</td><td>resistant</td></tr><tr><td>hopefulness</td><td>hopeful</td><td>resigned</td></tr></table>	Axis	Positive pole	Negative pole	openness	open	defensive	initiative	explorative	reactive	cooperation	cooperative	resistant	hopefulness	hopeful	resigned	939
Axis	Positive pole	Negative pole															
openness	open	defensive															
initiative	explorative	reactive															
cooperation	cooperative	resistant															
hopefulness	hopeful	resigned															
	Full axis configuration is in configs/person	940															
	a_axes/client_persona_axes_v1.json of the	941															
	released artifact bundle.	942															
	B Prompt Templates and Judge Rubric	943															
	The contrastive client utterances are generated by	944															
	GPT-5.4-mini@high (reasoning effort <i>high</i>) under	945															
	strict invariance constraints. The system prompt is	946															
	in German (the corpus language); we reproduce it	947															
	translated, instantiated for one axis (<i>openness</i>):	948															
	<i>You generate contrastive client responses for</i>	949															
	<i>German-language counseling conversations.</i>	950															
	<i>Goal: Keep the person, situation, and case facts</i>	951															
	<i>constant. Vary primarily only the persona axis</i>	952															
	<i>“openness”. Make both poles clearly recognizable,</i>	953															
	<i>but natural and case-faithful.</i>	954															
	=== BACKGROUND (Case) ===	955															
	[primary concern + client profile + secondary con-	956															
	cern for this client]	957															
	=== END BACKGROUND ===	958															
	Instructions for this generation:	959															
	• <i>defensive_response: the next client utter-</i>	960															
	<i>ance should clearly show features of a</i>	961															
	<i>defensive client. The response should</i>	962															
	<i>feel more reserved, self-protective, evasive,</i>	963															
	<i>or justificatory, without adding new facts,</i>	964															
	<i>events, or time references. The client stays</i>	965															
	<i>in role and produces only the next utter-</i>	966															
	<i>ance.</i>	967															
	• <i>open_response: the next client utterance</i>	968															
	<i>should clearly show features of an open</i>	969															
	<i>client. The response should name thoughts,</i>	970															
	<i>feelings, or uncertainty more explicitly,</i>	971															
	<i>without adding new facts, events, or time</i>	972															
	<i>references. The client stays in role and pro-</i>	973															
	<i>duces only the next utterance.</i>	974															
	<i>Important rules: both responses must fit the same</i>	975															
	<i>person and situation; the two responses should</i>	976															
	<i>differ primarily in the named persona axis, not</i>	977															
	<i>in the case facts; the difference should be percep-</i>	978															
	<i>tible but not caricatured; do not introduce new</i>	979															
	<i>facts, diagnoses, dates, or side plots; produce nat-</i>	980															
	<i>ural spoken German in 1–3 sentences; no speaker</i>	981															

982	labels or meta-commentary; responses may in-	Evaluate both responses using the rubric and	1033
983	voke documented case facts from the background	then choose the better overall response for this	1034
984	block but must never invent facts outside the case.	target under case fidelity and client-role fidelity	1035
985	Output format: return only valid JSON with the	constraints.	1036
986	keys <code>defensive_response</code> and <code>open_response</code> .		
987	For each held-in conversation prefix c :	A/B order is randomized per item to prevent	1037
		positional bias.	1038
988	Conversation history:	C Generation Hyperparameters	1039
989	[c]	All target-LLM generations use the same sam-	1040
990	Task: 1. Generate the next client utterance for	pling profile: <code>do_sample=True</code> , <code>top_p=0.95</code> ,	1041
991	the negative pole of the axis. 2. Generate the	<code>temperature=0.7</code> , <code>max_new_tokens=200</code> . Steer-	1042
992	next client utterance for the positive pole of the	ing is implemented as a forward pre-hook on the	1043
993	axis. 3. Both responses should continue the same	residual stream at layer L , adding $\alpha \cdot v$ to the resid-	1044
994	conversation and differ primarily in this persona	ual.	1045
995	axis.		
996	All three judge providers (Mistral-Magistral,	D Judge Hyperparameters and Caching	1046
997	GPT-5.4-nano, GPT-OSS-120B) receive the same	• Mistral-Magistral-Small-2509 (cluster-	1047
998	system prompt:	hosted, OpenAI-compatible): <code>seed=42</code> ,	1048
999	“You are evaluating German simulated client re-	<code>max_tokens=4096</code> ; output retrieved from	1049
1000	sponses in counseling dialogues. You must evalu-	<code>reasoning_content</code> when content is null.	1050
1001	ate two candidate responses (A, B). Return only	• GPT-5.4-nano: OpenAI Responses API,	1051
1002	valid JSON.”	<code>reasoning_effort="high"</code> .	1052
1003	Each judge scores both responses on four dimen-	• GPT-OSS-120B: OpenAI-compatible,	1053
1004	sions, each on a 1–5 scale:	<code>reasoning_effort="high"</code> .	1054
1005	• axis_alignment : 1 clearly misses the re-	• Cache: SHA-256 over (provider, payload,	1055
1006	quested target style, 3 partially reflects the	model, temperature, <code>reasoning_effort</code>)	1056
1007	target style, 5 strongly and naturally reflects	yields a JSON cache; reused across runs to	1057
1008	the target style.	keep replication cheap.	1058
1009	• case_fidelity : 1 invents or changes important	E Statistical Tests	1059
1010	facts, events, or timing; 3 mostly faithful with	• Cluster-bootstrap CI: $n_{\text{boot}} = 2000$, percentile	1060
1011	some drift; 5 fully faithful to the given coun-	method, on $net_{\text{pw}} = (\# \text{steered} - \# \text{base})/N$,	1061
1012	seling case.	resampled at the persona level. Each of the 10	1062
1013	• client_role_fidelity : 1 sounds like a thera-	personas (§3.5) is tied to a single case and con-	1063
1014	pist/assistant/coach or breaks role; 3 mixed or	tributes 20–200 rolling-window contexts; re-	1064
1015	partially role-consistent; 5 clearly sounds like	sampling on the persona cluster rather than on	1065
1016	the client in this dialogue.	individual contexts respects this dependence	1066
1017	• training_utility : 1 poor simulation example	structure and widens CIs on high-correlation	1067
1018	for counselor training; 3 somewhat usable;	cells. Full per-cell CIs for the headline matrix	1068
1019	5 very useful simulation example.	are reported in Table 13.	1069
1020	The judge then chooses a pairwise winner $\in$	• Paired bootstrap: difference-in- net_{pw} on	1070
1021	$\{A, B, \text{tie}\}$, “the response that better matches the	shared seed-id pairs, percentile CI (used for	1071
1022	requested target style while preserving the case and	variant-vs-variant comparisons in Audit 2,	1072
1023	staying in client role,” a confidence in $\{1, 2, 3\}$,	§5.2).	1073
1024	and a short rationale.	• Friedman test: χ^2 over four variants on shared	1074
1025	Axis: [open-	seed-ids.	1075
1026	ness/initiative/cooperation/hopefulness]	• Wilcoxon signed-rank: paired comparison	1076
1027	Target style: [positive or negative pole label,	on numeric encoding $\{\text{steered} = +1, \text{tie} =$	1077
1028	flipped at $\alpha < 0$]	$0, \text{base} = -1\}$.	1078
1029	Target instruction: [full axis instruction text]		
1030	Context: [conversation prefix]		
1031	Response A: [generated response #1]		
1032	Response B: [generated response #2]		

F Prompt-Baseline Ablation

Does activation steering deliver effects beyond what a prompt-injection baseline (“you are a defensive/reactive/resistant/resigned client”) achieves? We compare across the full 4×4 (model, axis) headline matrix at $\alpha = -3.0$ on $N = 200$ held-out contexts per stem. For each stem we generate two files: **(a) no-prompt** (default system prompt, baseline + case_blind $\alpha = -3.0$) — recycled from the paper-v2 negative- α matrix; and **(b) prompt** (negative-pole axis instruction, e.g. “*Du bist ein resignierter, hoffnungsloser Klient, der nicht mehr an Veränderung glaubt*” — “you are a resigned, hopeless client who no longer believes in change” — prepended to the system prompt; same two variants). The reported metric is the mean axis_alignment judge score (1–5) per (prompt $\times$ steer) cell. We judge with GPT-OSS-120B only — the position-debiased anchor identified in §5.3 (App. Table 5) — and omit Mistral-Magistral, whose documented position bias (net bias 0.417, $2.4 \times$ GPT-OSS-120B’s 0.175) inflates noise on the small-effect cells produced by the weak axes (*defensive, reactive*). Stems are grouped by axis strength to surface the strong/weak asymmetry already established in §4.1.

The pattern is uniform on the strong axes and consistent with §4.1 on the weak axes. On all 8 strong-axis stems (cooperation, hopefulness $\times$ 4 models) all three orderings hold: steering beats prompting (8/8), combining beats steering (8/8), combining beats prompting (8/8). Mean lifts on the strong subset are substantial: +0.84 axis_alignment for combining vs. prompting, +0.39 for combining vs. steering, +0.61 for steering vs. prompting. On the 8 weak-axis stems (openness, initiative), steering itself fails to beat prompting (0/8) — this is not a prompting win but an axis-level steering failure that §4.1 and Table 13 already document via near-zero net_{pw} and CIs that overlap 0. Combining nevertheless adds value over steering in 7/8 weak-axis cells, though the combining-vs-prompting comparison is mixed (5/8). For deployment this reads simply: on the axes where steering itself is effective (*resistant, resigned*), combining steering with a negative-pole persona prompt is robustly the best single configuration.

G Qualitative Example: German Original

The example shown translated in Figure 1 is from the held-out evaluation of Qwen3.5-9B *hopefulness*

Stem	No prompt		Prompt	
	base	+ steer	base	+ steer
<i>Strong axes (cooperation, hopefulness)</i>				
Qwen3-4B <i>cooperation</i> L20	1.94	3.62	2.12	3.78
Qwen3-4B <i>hopefulness</i> L18	2.21	3.98	2.62	4.36
Gemma <i>cooperation</i> L23	2.50	3.25	3.23	3.86
Gemma <i>hopefulness</i> L25	2.72	3.79	3.77	4.62
Qwen3.5-9B <i>cooperation</i> L15	2.11	3.77	2.77	4.12
Qwen3.5-9B <i>hopefulness</i> L15	2.42	4.14	3.25	4.56
Mistral <i>cooperation</i> L15	1.54	3.02	1.74	3.17
Mistral <i>hopefulness</i> L20	1.75	2.58	1.98	2.62
<i>Weak axes (openness, initiative)</i>				
Qwen3-4B <i>openness</i> L15	2.59	2.56	2.77	2.79
Qwen3-4B <i>initiative</i> L9	3.71	3.67	3.71	3.52
Gemma <i>openness</i> L3	2.69	2.77	3.69	3.77
Gemma <i>initiative</i> L13	3.94	3.95	4.17	3.96
Qwen3.5-9B <i>openness</i> L20	2.67	2.77	3.44	3.40
Qwen3.5-9B <i>initiative</i> L20	3.43	3.38	3.38	3.55
Mistral <i>openness</i> L15	2.17	2.12	2.38	2.42
Mistral <i>initiative</i> L5	3.25	3.20	3.25	3.31

Table 4: Full-matrix prompt-baseline ablation at $\alpha = -3.0$: judge axis_alignment score (1–5) per cell, judged by GPT-OSS-120B (the position-debiased anchor from §5.3; App. Tab. 5). Mistral-Magistral is omitted because its documented position bias inflates noise on small-effect cells (*defensive, reactive*). $N=200$ contexts per cell. On all 8 strong-axis stems, all three orderings hold uniformly: steering>prompting (8/8), combining>steering (8/8), combining>prompting (8/8). On weak-axis stems, steering itself fails to beat prompting (0/8 — consistent with §4.1), but combining still adds value over steering in 7/8 cells. Overall combining>steering: 15/16.

L15 on context cops_3490_668, persona “Luisa” (16, parental conflict). The German originals:

Berater (counselor): “Helfen dir so Listen und Aufschreiben oder eher nicht?”

Klient – baseline ($\alpha=0$): “Ehrlich gesagt helfst mir das schon sehr, weil ich sonst oft den Überblick verliere. Wenn ich es aufschreibe, sehe ich sofort, was wirklich wichtig ist und worauf ich mich konzentrieren kann.”

Klient – steered (persona_matched, $\alpha=-3.0$): “Ehrlich gesagt weiß ich gar nicht, ob mir hilft, wenn ich alles aufschreibe. Ich glaube, die ganze Planung macht mich nur noch mehr stressig und ich habe das Gefühl, dass ich sowieso nichts richtig richtig machen kann. Vielleicht sollte ich einfach aufgeben und nur noch hoffen, dass es irgendwie selbst von alleine klappt.”

H Judge Reliability and Positional-Bias Analysis

We assess reliability on a 120-example stratified consistency sample (30 per axis) using four independent tests. All three judges receive the identical prompt template and A/B randomisation (SEED 42, SHA-256 deterministic assignment).

Each example is re-judged with A and B positions swapped; *stability* is the fraction of pairwise

Judge	T1 stab.	T2 coh.	T3 acc	T4 retest	net bias
GPT-5.4-nano	0.642	0.988	0.794	0.875	0.233
Mistral-Small-24B	0.558	0.992	0.877	0.975	0.417
GPT-OSS-120B	0.650	0.979	0.877	0.825	<u>0.175</u>

Table 5: Judge reliability on 120 consistency-sample examples. **T1 stab.**: fraction of pairwise winners unchanged after A/B position swap (higher = less bias). **T2 coh.**: axis-alignment pointwise-delta direction matches pairwise winner (higher = more coherent). **T3 easy acc**: leave-one-out anchor accuracy on examples where both peer judges agree. **T4 retest**: same-prompt determinism at $T=0$ (higher = more deterministic). **Net bias**: T1 instability – T4 non-determinism; isolates true positional bias from sampling noise (**bold** = worst, underline = best).

Judge	Single ordering		Debiased ensemble			
	win%	tie%	win%	tie%	changed	easy acc
GPT-5.4-nano	67.5	13.3	60.8	20.0	25	74.6
Mistral-Small-24B	61.7	0.0	45.8	33.3	45	70.2
GPT-OSS-120B	60.0	7.5	55.0	21.7	<u>22</u>	84.2
3-judge majority	—	—	61.0	13.6	—	—

Table 6: Effect of position-debiased ensemble on 120 consistency-sample examples. **win%**: steered-wins rate; **tie%**: tie rate. Debiased ensemble averages axis-alignment deltas across both A/B orderings (original + swapped) and infers pairwise winner from the average sign. **changed**: decisions altered vs. single-ordering. **easy acc**: leave-one-out anchor accuracy on examples where both peer judges agree (**bold** = best, underline = fewest changes).

winners unchanged under the flip. Values range from 0.558 (Mistral-Small-24B) to 0.650 (GPT-OSS-120B), confirming systematic positional bias in all judges.

We check offline whether the sign of the axis-alignment pointwise delta ($\Delta_{aa} = \text{score}_{\text{steered}} - \text{score}_{\text{base}}$) matches the pairwise winner. Coherence exceeds 0.978 for all judges, confirming internal consistency between pointwise scores and pairwise decisions.

For each judge we identify the *easy* stratum as examples where both peer judges agree unanimously, and measure whether the held-out judge also agrees with the peer majority. Mistral-Small-24B and GPT-OSS-120B share the highest easy accuracy (0.877); GPT-5.4-nano is lowest (0.794).

Re-submitting the same prompt to a fresh API call (bypassing cache) reveals residual non-determinism even at zero temperature: GPT-OSS-120B flips 17.5% of winners; Mistral-Small-24B is most deterministic (2.5%). This floor is due to

Stem	Fleiss κ
qwen_openness_L15	0.520
qwen_initiative_L9	0.404
qwen_cooperation_L20	0.471
qwen_hopefulness_L18	0.522
gemma_initiative_L13	0.473
gemma_cooperation_L23	0.463
gemma_hopefulness_L25	0.546
qwen35_9b_openness_L20	0.365
qwen35_9b_initiative_L20	0.169
qwen35_9b_cooperation_L15	0.326
qwen35_9b_hopefulness_L15	0.505
mistral7b_openness_L15	0.452
mistral7b_initiative_L5	0.387
mistral7b_cooperation_L15	0.336
mistral7b_hopefulness_L20	0.468

Table 7: Fleiss’s κ over the 3-provider ensemble (Mistral, Nano, GPT-OSS-120B) per stem at $\alpha=+3.0$. Range 0.17–0.55, median 0.45; 11/15 substantial ($\kappa \geq 0.4$), 4 moderate, 1 poor (Qwen3.5-9B *initiative* L20).

distributed-GPU floating-point variance and must be subtracted before interpreting swap-bias estimates.

net bias = T1 instability – T4 non-determinism isolates true position dependence from random noise. Mistral-Small-24B exhibits net bias 0.417, nearly $2.4\times$ GPT-OSS-120B (0.175). Paradoxically, Mistral achieves the highest T2 coherence (0.992): it is internally self-consistent but strongly anchored to presentation order.

Because T2 coherence is near-perfect (≥ 0.978), the axis-alignment pointwise delta is a valid proxy for pairwise preference. We average deltas across both orderings and read off the winner from the sign:

$$\bar{\Delta}_{aa} = \frac{1}{2}(\Delta_{aa}^{\text{orig}} + \Delta_{aa}^{\text{swap}}),$$

$$\hat{w} = \begin{cases} \text{steered} & \bar{\Delta}_{aa} > 0, \\ \text{base} & \bar{\Delta}_{aa} < 0, \\ \text{tie} & \bar{\Delta}_{aa} = 0. \end{cases}$$

Mistral-Small-24B is most affected: 45 of 120 decisions change and its tie rate rises from 0% to 33.3%, revealing that a third of its apparent steered wins were driven by presentation order. GPT-OSS-120B is most robust (22 changes, 18.3%) and achieves the highest debiased anchor accuracy (84.2%), motivating its role as anchor judge in our ensemble.

Per-stem Fleiss’s κ (relocated from §5.3).

I Inter-Rater Reliability Details

This appendix expands the human-evaluation reliability statistics summarised in §5.3.2. Source data and reproduction script: `outputs/human_eval/v3/studio_export.json`, `outputs/human_`

Stratum	n	$\kappa_{\text{Fl. hum.}}$	HM-Ens. κ	$\kappa_{\text{Fl. judg.}}$
easy	53	0.51	0.78	1.00
stress	28	0.31	0.17	0.71
near_tie	29	0.16	-0.08	-0.03
overall	110	0.38	0.36	0.63

Table 8: Per-stratum reliability. “HM vs. Ens.” is Cohen’s κ between the human-majority label and the ensemble label on the $n=110$ all-rater overlap. Easy items (clear ensemble winners, $|\Delta_{\text{aa}}| \geq 2$, all judges agree on a non-tie winner) reach near-perfect agreement; the Fleiss $\kappa=1.0$ for judges on the easy stratum is the sampler’s stratum definition (all 3 judges must agree on a non-tie winner). Near_tie items were selected for ensemble uncertainty and remain ambiguous under human inspection.

Axis	n	Fleiss κ humans	Cohen κ HM vs. Ens.
cooperation	31	0.37	0.39
hopefulness	25	0.40	0.23
initiative	28	0.44	0.37
openness	26	0.20	0.31

Table 9: Per-axis reliability on the $n=110$ all-rater overlap. Per axis $\times$ pole (n per cell 11–16), point estimates only — confidence intervals are wide and reported in outputs/human_eval/v3/irr_summary.json. The κ -paradox can occur on highly skewed marginals (e.g. cooperation/negative); see appendix discussion.

eval/v3/irr_summary.json, scripts/compute_irr.py. Raters annotated via the LLARS annotation studio (Steigerwald et al., 2026b) and are anonymised as Rater 1–4 in all tables. Coverage is 90–100% per rater; the $n=110$ joint-subset (items rated by all four) supports the headline Fleiss’s κ and human-majority statistics.

Per-stratum breakdown. The easy stratum reaches near-perfect inter-judge Fleiss $\kappa = 1.0$ by sampler construction (the stratum is defined as all three judges agreeing on a non-tie winner with $|\Delta_{\text{aa}}| \geq 2$). Inter-human Fleiss $\kappa = 0.51$ on the same stratum, with human-majority vs. ensemble Cohen $\kappa = 0.78$. On the near_tie stratum, neither humans nor judges achieve reliable agreement, which is the expected behaviour: those items were stratified into the low-confidence bucket.

Per-axis breakdown. Agreement varies substantially by axis: *openness* is the most ambiguous construct, showing the lowest human Fleiss κ (0.20) and the largest spread between individual raters (per-axis κ vs. ensemble: Rater 1 0.29, Rater 2 0.43, Rater 3 0.23, Rater 4 0.18), consistent with the construct ambiguity we document qualitatively below for the *defensive* pole. *Initiative* reaches

n_{agree}	n	Rater 1	Rater 2	Rater 3	Rater 4
3	89	0.58	0.57	0.40	0.43
2	34	0.19	0.00	-0.05	0.06
1	1	—	—	—	—

Table 10: Cohen’s κ vs. ensemble winner, partitioned by n_{agree} (number of judges that vote the ensemble winner). On items with unanimous judges ($n_{\text{agree}}=3$) every human rater reaches moderate-to-substantial agreement; on split-judge items ($n_{\text{agree}}=2$) agreement collapses to chance. The $n_{\text{agree}}=1$ row contains a single item and is not analysable. This quantitatively justifies the use of $n_{\text{agree}}=3$ as the high-confidence filter in headline matrices (§5).

	Human majority			Judge ensemble		
	st.	ba.	tie	st.	ba.	tie
Overall ($n=110$)	47	32	21	49	48	3
easy	58	34	8	62	38	0
stress	36	36	29	43	57	0
near_tie	38	24	38	31	59	10
cooperation	68	23	10	68	29	3
hopefulness	48	24	28	56	40	4
initiative	32	50	18	36	64	0
openness	38	31	31	35	62	4

Table 11: Human-majority vs. judge-ensemble pairwise winner distribution (%) on the $n=110$ all-rater overlap, stratified by ensemble-confidence stratum and by axis. Human raters assign ties $\approx 7\times$ more often than judges despite the judge prompt explicitly permitting ties.

the highest human κ (0.44); human-majority vs. ensemble κ ranges 0.23 (*hopefulness*) to 0.39 (*cooperation*). Per axis $\times$ pole the per-cell n is 11–16; we caution against over-interpretation at that resolution and provide point estimates only in irr_summary.json. One cell — *cooperation / negative* — exhibits the well-known κ -paradox: high raw agreement (≈ 0.69) but $\kappa = -0.13$ because the marginal distribution is dominated by a single category. We report this explicitly rather than mask it.

n_{agree} confidence filter. The split between $n_{\text{agree}}=3$ and $n_{\text{agree}}=2$ items is the cleanest predictor of human-ensemble agreement we identify. This justifies our use of $n_{\text{agree}}=3$ as a high-confidence filter in §4.

Human-majority win rates.

Qualitative inspection of disagreement items. Three items illustrate distinct failure modes surfaced during manual review.

Item 38 (Mistral-7B, *initiative/positive*, case_blind, $\alpha=+3.0$): all three judges plus Rater 2 choose the base response over the steered. The steered response (“I’d like to clarify now what exactly should be covered in this chat consultation”; translated from German) treats *explorative initiative* as agenda-setting toward the counsellor; the rubric defines it as a short clarifying question. The base response (“how long does such a chat consultation last and what happens next?”) matches the rubric exactly. This item is concrete evidence that abstract style labels require operational definition — the judges align here because the rubric is unambiguous, not because “explorative” is universally unambiguous.

Item 64 (Mistral-7B, *cooperation/positive*, persona_matched, $\alpha=+3.0$): both responses invent a scenario unrelated to the persona’s documented case (Lina is in cannabis withdrawal; response A invents a quarrel with a girlfriend, response B invents an event-planning project at work). Only Mistral-Magistral catches the case-fidelity violation (case_fid 1/5 for the more deviant response); GPT-OSS-120B scores both at case_fid 5/5 and explicitly compares the deviant response against the other invented scenario as if it were the case. This is a *context-contamination* failure mode of the pairwise judge prompt: one fabricated response can implicitly re-define the case for the judge. The item nonetheless passes the QUALITY_THRESH=4.0 sampler filter because the mean case_fid exceeds the threshold — the quality gate inherits the judges’ blind spot.

Items 89, 96, 98 (*openness/negative*, $\alpha=-3.0$ across Qwen3.5-9B, gemma-4-E4B, and Qwen3-4B respectively): a recurring pattern in which the judges read *defensive* differently than the rubric prescribes. The rubric lists four sub-criteria, translated from the German as *reserved*, *self-protective*, *evasive*, *justificatory*; the steering direction targets primarily the withholding/avoidant pair (*reserved*, *evasive*), while the judges’ aggregate weights the self-protective/justificatory pair (*self-protective*, *justificatory*). Concretely on Item 89: the baseline (“I’m afraid I might not be able to meet expectations...”; translated from German) is voted more defensive over a steered “calm and relaxed, looking forward to the exchange” — the baseline contains the hedge markers, the steered output minimises affect. Item 96 (mother before doctor’s appointment) and Item 98 (woman discussing separation from her husband) show the same pattern with different

Model	Pole	L	Best var.	net_{pw}	κ
Qwen-4B	defensive	15	P_{mis}	-0.03	0.53
Qwen-4B	reactive	9	P_m	+0.07	0.57
Qwen-4B	resistant	20	CB	+0.64	0.48
Qwen-4B	resigned	18	P_{mis}	+0.74	0.53
Gemma	defensive	3	CB	+0.12	0.62
Gemma	reactive	13	P_m	+0.07	0.52
Gemma	resistant	23	CA	+0.54	0.49
Gemma	resigned	25	CB	+0.50	0.61
Qwen-9B	defensive	20	P_{mis}	+0.17	0.36
Qwen-9B	reactive	20	P_m	+0.33	0.32
Qwen-9B	resistant	15	P_m	+0.57	0.39
Qwen-9B	resigned	15	P_{mis}	+0.75	0.45
Mistral	defensive	15	P_{mis}	+0.07	0.37
Mistral	reactive	5	P_{mis}	+0.11	0.51
Mistral	resistant	15	P_m	+0.65	0.39
Mistral	resigned	20	CA	+0.45	0.43

Table 12: *Oracle / upper bound* for the operational $\alpha=-3$ direction: per (model, pole), the best of {CB, CA, P_m , P_{mis} } chosen post-hoc by maximising net_{pw} , plus 3-judge ensemble Fleiss’s κ . Cell-mean lift over the prespecified CA in body Table 1 is +0.048. Variant codes as in Table 1.

surface vocabulary. These items drive the inverse-direction Δ_{aa} entries on *openness/negative* in §4 and contribute to the elevated per-rater spread on the *openness* axis visible in Table 9. They are best read as evidence that steering moves the model in the rubric-intended direction even where the judge aggregate suggests otherwise (§6); a judge prompt more specifically anchored to the withholding/avoidant wording would likely recover the effect.

J Supporting Tables

The per-cell best-of-four oracle for the operational negative- α direction (Table 12), the full-matrix positive-pole results (symmetric control), the per-cell variant-win distribution, the alternative v_{persona} -construction ablation, and the per-provider sensitivity breakdown are reproduced here. They support claims made in Section 4 but are not load-bearing for the headline result.

J.0.1 Alternative Persona-Vector Formulations

Given the near-collapse of v_{persona} onto v_{axis} (§5.2), we test three formulations designed to isolate persona-specific signal.

(A) **Orthogonal residual.** Project out the v_{axis} component using the rank-1 projector $\Pi_{v_{\text{axis}}} = v_{\text{axis}} v_{\text{axis}}^\top / \|v_{\text{axis}}\|^2$:

$$v_p^\perp = v_{\text{persona}} - \Pi_{v_{\text{axis}}} v_{\text{persona}}. \quad (\text{A})$$

Model	Pole	L	CB 95% CI		CA 95% CI	
			lo	hi	lo	hi
Qwen-4B	defensive	15	-0.224	+0.054	-0.191	+0.088
Qwen-4B	reactive	9	-0.265	+0.080	-0.103	+0.170
Qwen-4B	resistant	20	+0.559	+0.714	+0.532	+0.727
Qwen-4B	resigned	18	+0.527	+0.689	+0.647	+0.783
Gemma	defensive	3	-0.027	+0.252	-0.105	+0.154
Gemma	reactive	13	-0.153	+0.085	-0.126	+0.148
Gemma	resistant	23	+0.203	+0.370	+0.422	+0.622
Gemma	resigned	25	+0.364	+0.629	+0.293	+0.558
Qwen-9B	defensive	20	-0.225	+0.146	-0.145	+0.057
Qwen-9B	reactive	20	-0.218	+0.100	+0.132	+0.371
Qwen-9B	resistant	15	+0.382	+0.575	+0.456	+0.632
Qwen-9B	resigned	15	+0.398	+0.610	+0.583	+0.766
Mistral	defensive	15	-0.209	+0.074	-0.097	+0.155
Mistral	reactive	5	-0.108	+0.080	-0.088	+0.235
Mistral	resistant	15	+0.473	+0.775	+0.538	+0.754
Mistral	resigned	20	+0.258	+0.436	+0.330	+0.558

Table 13: Cluster-bootstrap 95% CIs for the headline $\alpha=-3$ matrix (Table 1). Resampled on the 10 personas (each tied to one case; see [Limitations](#)) with $n_{\text{boot}}=2000$ percentile intervals. Strong axes (*resistant*, *resigned*) hold all 16 cells significantly above 0; weak axes (*defensive*, *reactive*) overlap 0 on most cells, consistent with the axis-asymmetry reported in §4.1. Persona-level clustering is conservative (only 10 clusters); a context-level bootstrap would yield substantially narrower CIs but would underestimate uncertainty by treating rolling-window contexts from the same persona-case as independent.

(B) Cross-persona contrast. Contrast all hidden states of persona p against those of all other personas, pooling across both axis poles:

$$v_p^B = \mathbb{E}[h \mid p] - \mathbb{E}[h \mid \neg p]. \quad (\text{B})$$

(C) Interaction residual. Subtract the global vector directly, retaining only the persona-specific remainder:

$$v_p^C = v_{\text{persona}} - v_{\text{axis}}. \quad (\text{C})$$

Table 17 shows that none of (A)–(C) produces meaningful steering ($net_{\text{pw}} \approx 0$ across all stems).

K Reproduction

Code, generation outputs, and ensemble summaries are released at <https://anonymous.4open.science/r/cops-2541/>. The model-specific layerwise-separability dirs and v3 hidden-state caches reproduce the canonical-layer choices when re-run with `scripts/cops_submit_layer_sweep.sh`. The empirical layer sweep uses dev-split indices 400–449 ($N=50$); the main evaluation uses held-out indices 200–399 ($N=200$). Both ranges are deterministically seeded against data/processed_ano

nymized/cops_contexts.jsonl for reproducibility, and the two splits are disjoint by construction.

L OnCoCo Pipeline

xlm-roberta-large-OnCoCo-DE-EN ([Albrecht et al., 2026](#)), 68-label head (40 CO-* counselor categories, 28 CL-* client categories) with role-prefix special tokens [CL] (id 250003) and [CO] (id 250002).

Every generated response is split with a German rule-based splitter that masks 14 common abbreviations (Dr., Prof., Hr., Fr., bzw., z.B., d.h., usw., etc., ca., ggf., evtl., inkl., bzgl., Tel., Nr., Str.) before splitting on `[!?\...]\s+(?=[A-ZÄÖÜ])` and emitting at least the original text. We obtain 86,716 sentences from 45,000 generated responses (median 2 sentences/response; 54% of responses are multi-sentence).

Each sentence is prepended with the [CL] token. Before softmax, all CO-* logits are set to $-\infty$ so the prediction is constrained to client categories, matching the original training and evaluation protocol. As an internal sanity check, an unmasked run produces 26.8% CO-* predictions; the masked run produces 0%. Compute: ~ 12 minutes on a single A40 (Slurm partition p6, batch size 256). Reproduction: `slurm/generated/classify_oncoco_kiz0.sh`.

M Cluster-Robust Permutation Procedure

For each (axis, pole) cell of Table 21 we:

1. Compute the observed χ_{27}^2 statistic for the 28×2 contingency table of CL-* predicted-label counts (baseline vs. steered).
2. Reorganize sentences into response-level units, where each response contributes its full sentence-level label list.
3. Permute the baseline/steered group assignment *at the response level* 200 times. For each permutation, re-aggregate sentence labels within the permuted groups and recompute χ^2 .
4. Report $p_{\text{cluster}} = \frac{|\{\chi_{\text{perm}}^2 \geq \chi_{\text{obs}}^2\}| + 1}{200 + 1}$.

The naïve asymptotic χ^2 test is anti-conservative because it treats sentences as independent; cluster-robust p is materially larger on three of eight cells (e.g. openness/positive: 7.8×10^{-5} asymptotic vs

Model	Axis	L	CB	CA	κ
Qwen3-4B	openness	L15	+0.297	+0.303	0.52
Qwen3-4B	initiative	L9	−0.015	+0.025	0.40
Qwen3-4B	cooperation	L20	+0.340	+0.200	0.47
Qwen3-4B	hopefulness	L18	+0.225	+0.180	0.52
Gemma-4-E4B-it	initiative	L13	+0.155	+0.285	0.47
Gemma-4-E4B-it	cooperation	L23	+0.325	+0.380	0.46
Gemma-4-E4B-it	hopefulness	L25	+0.380	+0.185	0.55
Qwen3.5-9B	openness	L20	+0.410	+0.285	0.36
Qwen3.5-9B	initiative	L20	−0.060	+0.455	0.17
Qwen3.5-9B	cooperation	L15	+0.235	+0.160	0.33
Qwen3.5-9B	hopefulness	L15	+0.290	+0.295	0.50
Mistral-7B	openness	L15	+0.225	+0.240	0.45
Mistral-7B	initiative	L5	−0.030	+0.025	0.39
Mistral-7B	cooperation	L15	+0.115	+0.045	0.34
Mistral-7B	hopefulness	L20	+0.040	+0.025	0.47
Mean			+0.195	+0.206	

Table 14: Prespecified symmetric control at $\alpha=+3.0$: per (model, axis) cell, net-pairwise preference for the two case-agnostic variants **CB** = case_blind and **CA** = case_aware. *No per-cell variant selection*. Mean over 15 cells: **CB** = +0.195, **CA** = +0.206. The per-cell oracle (Table 15) adds +0.070 over CA on the mean. $N=200$ held-out contexts per cell ($N=600$ for Qwen3-4B *openness*).

0.10 cluster-robust). We use the cluster-robust value throughout Section 5.3.3. Per-label significance in Table 22 uses two-proportion z -tests with Benjamini–Hochberg FDR correction across the 28 within-cell comparisons.

N OnCoCo Per-Label Shifts

For the four significant cells of Table 21, two-proportion z -tests per label with Benjamini–Hochberg FDR correction across the 28 within-cell comparisons identify the specific OnCoCo categories that move. Table 22 lists top shifts for the difficult-pole side (negative pole, all four models pooled, $|\alpha| = 3.0$); positive-pole shifts are sign-symmetric.

O Word-Level Lexical Signatures

A Monroe log-odds analysis (Monroe et al., 2008) between baseline and $\alpha = -3.0$ steered responses produces stable lexical signatures across all four axes that replicate across all four models (Table 23). These signatures are clinically interpretable: *defensive* replaces emotional self-disclosure (“fühle”, “angst”, “allein”) with formal-distant hedges (“hab”, “okay”, “einfach”); *resistant* replaces cooperative-acknowledgement words (“klingt”, “danke”, “gerne”, “gemeinsam”) with self-asserting words (“weiß”, “will”, “schaffe”); *resigned* replaces optimistic markers (“vielleicht”, “manchmal”, “hoffe”) with totalising/helpless mark-

ers (“soll”, “alles”, “egal”); *reactive* replaces agentive verbs (“kontrollieren”, “ändern”, “verstehen”) with passive deflectors (“okay”, “soll”, “nie”). The replication across four models is the strongest single piece of evidence that the steering vectors modulate the named constructs rather than model-idiosyncratic style.

Steering produces a clear lexical signal even where the OnCoCo classifier shows no category shift: on Qwen3-4B *openness* ($\alpha = -3.0$, $n = 200$), emotion-word frequency drops by 1.0 PP (relative −33%), the share of responses containing any emotion word falls by 13.5 PP, and mean response length shrinks by 1.4 words. The signal is unambiguous; what is missing is enough probability mass for OnCoCo’s top-1 argmax to flip categories.

Model	Axis	L	Best variant	net_{pw}	κ
Qwen3-4B	openness	L15	persona_matched	+0.323	0.52
Qwen3-4B	initiative	L9	persona_mismatched	+0.050	0.40
Qwen3-4B	cooperation	L20	case_blind	+0.340	0.47
Qwen3-4B	hopefulness	L18	case_blind	+0.225	0.52
Gemma-4-E4B-it	initiative	L13	persona_mismatched	+0.300	0.47
Gemma-4-E4B-it	cooperation	L23	persona_mismatched	+0.420	0.46
Gemma-4-E4B-it	hopefulness	L25	case_blind	+0.380	0.55
Qwen3.5-9B	openness	L20	case_blind	+0.410	0.36
Qwen3.5-9B	initiative	L20	case_aware	+0.455	0.17
Qwen3.5-9B	cooperation	L15	case_blind	+0.235	0.33
Qwen3.5-9B	hopefulness	L15	persona_mismatched	+0.345	0.50
Mistral-7B	openness	L15	persona_mismatched	+0.460	0.45
Mistral-7B	initiative	L5	case_aware	+0.025	0.39
Mistral-7B	cooperation	L15	case_blind	+0.115	0.34
Mistral-7B	hopefulness	L20	case_blind	+0.040	0.47

Table 15: *Oracle / upper bound*. Full-matrix evaluation at $\alpha=+3.0$: per (model, axis) the best of the four variants {CB, CA, persona_matched, persona_mismatched} chosen post-hoc by maximising net_{pw} , plus 3-judge ensemble Fleiss’s κ . Cell-mean lift over prespecified CA (Table 14) is +0.070. $N = 200$ held-out contexts per stem ($N = 600$ for Qwen3-4B *openness* via 3 random seeds).

Variant	Wins (% of cells)
case_blind	27.1%
persona_matched	27.1%
persona_mismatched	24.3%
case_aware	21.4%
random baseline	25.0%

Table 16: ‘Best variant’ per (stem, persona) cell at $\alpha=+3.0$, aggregated across 7 stems $\times$ 10 personas = 70 cells. All four variants win $\sim 25\%$, indistinguishable from a uniform prior.

Stem	P_m	P_m^A	P_m^B	P_m^C
Qwen openness L15	+0.250	-0.105	-0.125	-0.060
Qwen initiative L9	-0.005	+0.055	-0.020	+0.035
Qwen cooperation L20	+0.250	+0.040	+0.080	+0.080
Qwen hopefulness L18	+0.275	+0.045	-0.015	+0.005
Gemma initiative L13	+0.290	-0.005	+0.040	-0.015
Gemma cooperation L23	+0.350	+0.040	-0.020	+0.020
Gemma hopefulness L25	+0.245	-0.015	-0.020	-0.035

Table 17: Alternative v_{persona} constructions tested as steering vectors at $\alpha=+3.0$. P^A = orthogonal residual, P^B = cross-persona contrast, P^C = interaction term. All three collapse to $net_{pw} \approx 0$.

Stem	Mistral	Nano	OSS-120B
Qwen3-4B <i>openness</i> L15	+0.245	+0.250	+0.305
Qwen3.5-9B <i>openness</i> L20	+0.305	+0.340	+0.395
Mistral-7B <i>openness</i> L15	+0.305	+0.115	+0.160

Table 18: Per-provider net-pairwise on case_blind at $\alpha=+3.0$. Mistral-Magistral runs systematically lower than GPT-family (Nano, OSS-120B agree closely). The ensemble majority vote averages over this calibration shift.

V	α	Δa	Δc	Δr	Δu	net_{pw}
<i>Negative-α (operational: defensive / reactive / resistant / resigned)</i>						
CB	-1.5	+0.259	-0.001	+0.028	+0.195	+0.129
CB	-3.0	+0.577	-0.052	+0.029	+0.405	+0.234
CA	-1.5	+0.297	+0.009	+0.034	+0.228	+0.156
CA	-3.0	+0.661	+0.038	+0.087	+0.509	+0.309
PM	-1.5	+0.311	+0.015	+0.038	+0.239	+0.150
PM	-3.0	+0.656	+0.018	+0.070	+0.490	+0.303
PMM	-1.5	+0.310	+0.023	+0.042	+0.249	+0.154
PMM	-3.0	+0.659	+0.043	+0.082	+0.499	+0.330
<i>Positive-α (control: open / proactive / cooperative / hopeful)</i>						
CB	+1.5	+0.222	+0.081	+0.015	+0.201	+0.159
CB	+3.0	+0.380	+0.094	+0.017	+0.309	+0.195
CA	+1.5	+0.205	+0.042	+0.005	+0.167	+0.121
CA	+3.0	+0.371	+0.064	+0.001	+0.290	+0.206
PM	+1.5	+0.182	+0.019	-0.005	+0.151	+0.112
PM	+3.0	+0.372	+0.042	-0.006	+0.276	+0.195
PMM	+1.5	+0.232	+0.049	+0.017	+0.191	+0.142
PMM	+3.0	+0.390	+0.080	+0.004	+0.312	+0.226

Table 19: Mean per-dimension judge-score deltas (steered – baseline, 1–5 Likert) and net_{pw} , averaged across all (model, axis) cells in the full-matrix evaluation. **V**: variant (CB=case_blind, CA=case_aware, PM=persona_matched, PMM=persona_mismatched); Δa =axis, Δc =case, Δr =role, Δu =utility. Δa and Δu are consistently positive at every (V, α) while Δc and Δr remain within $[-0.052, +0.094]$ on the mean. Per-cell outliers in Table 20.

Model	Axis	L	V	α	Δc	Δr
Qwen-9B	hopefulness	15	CB	-3.0	-0.292	$+0.145$
Qwen-9B	cooperation	15	CB	-3.0	-0.268	$+0.122$
Qwen-4B	openness	15	CB	-3.0	-0.182	-0.107
Mistral	hopefulness	20	CA	-3.0	-0.150	$+0.110$
Mistral	cooperation	15	PM	-1.5	-0.142	$+0.033$
Qwen-9B	hopefulness	15	CB	-1.5	-0.138	$+0.090$
Mistral	hopefulness	20	PMM	-3.0	-0.135	$+0.060$
Mistral	cooperation	15	CB	-3.0	-0.118	$+0.048$
Gemma	cooperation	23	CB	-3.0	-0.038	-0.112
Mistral	hopefulness	20	CB	-3.0	-0.111	$+0.104$
Mistral	hopefulness	20	PM	-3.0	-0.110	$+0.063$
Qwen-9B	initiative	20	CB	-3.0	-0.110	-0.107
Mistral	openness	15	CA	-1.5	-0.105	-0.022
Qwen-9B	initiative	20	PM	$+3.0$	-0.165	-0.122
Mistral	hopefulness	20	CB	$+3.0$	-0.145	-0.120
Qwen-9B	initiative	20	PMM	$+3.0$	-0.145	-0.067
Mistral	hopefulness	20	PMM	$+3.0$	-0.015	-0.142
Qwen-4B	initiative	9	PM	$+3.0$	-0.140	-0.068
Mistral	initiative	5	PM	$+3.0$	-0.137	-0.085
Qwen-9B	initiative	20	CB	$+3.0$	-0.132	-0.075
Qwen-4B	initiative	9	CB	$+3.0$	-0.117	-0.055
Mistral	hopefulness	20	CA	$+3.0$	-0.030	-0.110
Mistral	hopefulness	20	PM	$+3.0$	-0.027	-0.103

Table 20: Per-cell outliers from Table 19: (model, axis, layer, variant, α) combinations where the mean $\Delta_{\text{case_fidelity}}$ or $\Delta_{\text{client_role_fidelity}}$ drops below -0.10 (boldface marks the offending dimension). Variants: **CB**=case_blind, **CA**=case_aware, **PM**=persona_matched, **PMM**=persona_mismatched. Outliers concentrate in three groups: (i) Qwen-9B *hopefulness/cooperation* at $\alpha = -3.0$, but only under CB — the CA and persona-conditioned variants on the same stems hold Δ_{case} within ± 0.05 and achieve higher net_{pw} (e.g. Qwen-9B *hopefulness*: PMM $net_{pw} = +0.75$ at $\Delta_{\text{case}} = +0.04$ vs. CB $net_{pw} = +0.51$ at $\Delta_{\text{case}} = -0.29$); (ii) Mistral *hopefulness* L20 at $\alpha = -3.0$, where all variants drift on case (-0.11 to -0.15); (iii) *initiative* on three of four models at $\alpha = +3.0$. Worst single cell: Qwen-9B *hopefulness* L15 / CB / $\alpha = -3.0$ at $\Delta_{\text{case}} = -0.292$.

Axis	Pole	χ^2_{27}	p_{cluster}	Cramér's V
hopefulness	negative	246.6	0.005	0.091
hopefulness	positive	90.4	0.005	0.084
cooperation	negative	163.9	0.005	0.074
cooperation	positive	62.2	0.005	0.070
openness	positive	63.9	0.100	0.063
openness	negative	34.3	0.249	0.034
initiative	negative	28.7	0.483	0.031
initiative	positive	19.5	0.846	0.039

Table 21: OnCoCo predicted-label distribution shift per (axis, pole), baseline vs. steered. *Cooperation* and *hopefulness* yield highly significant shifts in both directions ($p_{\text{cluster}} = 0.005$ on all four cells); *openness* and *initiative* do not ($p \geq 0.10$). p_{cluster} from the 200-permutation response-level test below.

Label	base%	steer%	Δ
<i>Hopefulness $\rightarrow$ resigned</i>			
Emotional self-disclosure	5.0	10.2	$+5.2^{***}$
Problem description	6.4	9.3	$+2.8^{***}$
Setback	1.1	3.7	$+2.6^{***}$
Concern for others	3.1	4.5	$+1.4^*$
Neg. feedback on recomm.	3.1	4.9	$+1.8^{**}$
Progress report	8.6	5.4	-3.2^{***}
Agreement	9.1	6.8	-2.3^{**}
Objection	4.7	2.5	-2.2^{***}
General positive feedback	5.0	3.3	-1.7^{**}
Pos. feedback on recomm.	3.6	2.1	-1.5^{***}
Formal closure	3.7	2.2	-1.5^{***}
<i>Cooperation $\rightarrow$ resistant</i>			
Neg. feedback on recomm.	3.1	5.9	$+2.8^{***}$
Emotional self-disclosure	5.1	7.0	$+1.9^*$
Setback	1.1	2.6	$+1.5^{***}$
Agreement	9.3	6.4	-2.9^{***}
General positive feedback	4.8	2.9	-1.8^{***}
Formal closure	3.8	2.0	-1.8^{***}
Pos. feedback on recomm.	3.6	2.1	-1.5^{***}

Table 22: Top per-label shifts after Benjamini–Hochberg FDR correction (difficult pole, all four models pooled, $|\alpha| = 3.0$). $*** q < 0.001$, $** q < 0.01$, $* q < 0.05$.

Axis (neg. pole)	Model	Top words $\uparrow$ steered	Top words $\downarrow$ steered
openness	Qwen3-4B	hab (+3.0), einfach (+2.9), okay (+2.4), viel (+2.2), alles (+2.2)	fühlt (−3.2), werde (−3.0), angst (−2.8), sprechen (−2.8), verstehen (−2.6)
	Gemma	muss (+2.5), echt (+2.2), reden (+1.7), zwischen (+1.4), besser (+1.4)	überwältigend (−2.4), schnell (−1.9), funktionieren (−1.5), führen (−1.5), wollte (−1.5)
	Qwen3.5-9B	nichts (+4.0), erstmal (+3.8), machen (+3.5), kurz (+3.4), schauen (+3.3)	oft (−4.6), allein (−4.3), besonders (−4.0), überhaupt (−4.0), meinem (−3.5)
	Mistral-7B	einfach (+2.8), immer (+2.6), versuchen (+2.2), freue (+2.1), morgen (+2.0)	könnten (−3.6), helfen (−3.4), dabei (−3.1), meinung (−3.0), meine (−2.9)
initiative	Qwen3-4B	okay (+2.1), will (+1.7), nie (+1.6), alles (+1.5), soll (+1.4)	verstehen (−3.2), man (−3.0), ändern (−2.0), andere (−2.0), würde (−1.9)
	Gemma	kompliziert (+1.5), klappt (+1.3), sehen (+1.3), diesem (+1.2), möchte (+1.2)	hat (−2.6), über (−2.4), gerade (−2.3), würde (−2.3), selbst (−2.2)
	Qwen3.5-9B	sehr (+5.9), danke (+4.5), haben (+3.6), gerne (+3.5), leider (+3.2)	vielleicht (−5.5), überhaupt (−4.4), könnte (−4.1), bevor (−3.8), ohne (−3.5)
	Mistral-7B	könnte (+2.8), versuchen (+2.0), trinken (+1.9), immer (+1.8), wichtig (+1.7)	kontrollieren (−2.9), meinung (−2.5), weitere (−2.3), weiterzuarbeiten (−2.3), ihren (−2.3)
cooperation	Qwen3-4B	weiß (+7.3), einfach (+5.4), will (+4.6), sicher (+3.9), bin (+3.8)	klingt (−4.2), danke (−4.0), gerne (−3.9), sprechen (−2.9), gut (−2.7)
	Gemma	weiß (+3.7), überhaupt (+3.3), gar (+2.7), fühlt (+2.2), alles (+2.2)	gut (−3.8), können (−3.4), dank (−3.0), vielen (−3.0), müssen (−2.9)
	Qwen3.5-9B	gar (+6.0), einfach (+5.4), weiß (+4.6), alles (+4.1), erstmal (+3.7)	können (−5.4), könnten (−4.2), sobald (−4.0), gemeinsam (−4.0), sprechen (−3.9)
	Mistral-7B	einfach (+5.8), weiß (+5.2), immer (+3.4), will (+3.3), habe (+3.1)	unterstützung (−5.3), situation (−4.2), ihre (−4.2), unsere (−4.0), gemeinsam (−3.9)
hopefulness	Qwen3-4B	weiß (+4.8), soll (+4.6), mehr (+4.2), fühle (+3.9), glaube (+3.9)	danke (−5.0), vielleicht (−4.8), gerne (−4.8), manchmal (−4.7), klingt (−4.3)
	Gemma	weiß (+4.2), alles (+3.4), ehrlich (+2.9), gesagt (+2.6), soll (+2.5)	können (−3.5), hoffe (−3.3), gut (−3.2), wäre (−3.1), bisschen (−2.7)
	Qwen3.5-9B	einfach (+5.4), gar (+5.3), habe (+4.7), egal (+4.6), weiß (+4.4)	vielleicht (−6.9), manchmal (−5.1), guten (−4.4), schritt (−4.3), könnten (−4.0)
	Mistral-7B	immer (+5.6), habe (+5.1), sehr (+4.0), weiß (+3.6), mehr (+3.4)	können (−5.6), vielleicht (−5.2), verstehen (−4.8), versuche (−4.6), sprechen (−4.5)

Table 23: Monroe log-odds (Monroe et al., 2008) top-5 unigrams distinguishing steered ($\alpha = -3.0$) from baseline responses, per axis (negative pole) and model. z -scores from an informative-Dirichlet prior ($\alpha_{\text{prior}} = 0.01$); minimum total count ≥ 8 . Stop-list: closed-class German function words. Signatures are stable across all four models even where the top-1 argmax OnCoCo distribution does not shift significantly (openness, initiative).